

# Mohit Dubey

Senior Full Stack Developer (React & Node.js) | AI Agents & LLM Applications

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## SUMMARY

Senior Full Stack Developer with 5+ years of experience designing and building high-performance **React** applications and **Python/Node.js** backend services, paired with deep expertise in **AI agent** and **LLM application** development end-to-end — including agentic orchestration logic (**LangChain**, **LangGraph**, **ReAct**-style reasoning/action loops). Skilled in **REST/GraphQL API integration**, agent tooling, and **observability/tracing** for production agent systems. Comfortable owning architecture end-to-end in **Agile/Scrum** teams, with hands-on experience on the **Databricks** platform.

## WORK EXPERIENCE

### Software Engineer — Publicis Sapient

June 2025 - Present

- Designed and deployed an enterprise-grade **GraphRAG**-based AI agent for a B2B SaaS platform end-to-end — agentic reasoning logic in **Python** using **LangGraph** and **DeepAgents** with **ReAct**-style reasoning/action loops, paired with a high-performance **React** UI — improving code retrieval accuracy by 38% and cutting developer resolution time by 60%.
- Built an **MCP server** with dynamic tool discovery, semantic search, and context orchestration to extend agent capabilities and integrate backend tools, reducing LLM context overhead by 70% and lowering token/inference costs across production workloads.
- Architected a **multi-tenant, multi-agent AI platform** with real-time guardrails, memory management, and **observability/tracing** across agent runs, enabling secure deployment across multiple tenants at sub-second latency.
- Orchestrated scalable agent workflows using **LangGraph**, vector databases, reranking, and hybrid retrieval; collaborated with backend teams to integrate **REST APIs** and services, increasing answer relevance by 42% and supporting thousands of concurrent enterprise requests.
- Worked in **Agile/Scrum** teams, partnering with executive leadership, Product, Engineering, and Data Governance to quantify ROI, enforce ethical AI/compliance standards, and drive new revenue streams from AI initiatives.
- Managed full project delivery from architecture design through production rollout, maintaining a consistent 95% project success rate.
- Built a **client-server UI version sync system** (React + Nginx) and a **reusable configurable table component** adopted across 15+ modules, improving consistency and reducing maintenance overhead.
- Centralized API interactions via a shared **Python/Node.js middleware** network layer with consistent error handling and retry logic, improving reliability across agent-to-backend data flow.
- Engineered scalable frontend architecture using **React.js**, **Next.js**, and micro-frontends — with code-splitting, lazy loading, and dynamic imports — improving platform performance by 60% and securing workflows with **OAuth2**.
- Implemented **test automation** and **regression testing** suites for **RAG** and LLM services in **CI/CD**, catching retrieval and response-quality regressions before release; integrated **Elasticsearch** for keyword and hybrid search over large document sets and deployed and debugged services on **Linux** environments.

### Software Engineer — Decode

Aug 2023 - May 2025

- Architected an enterprise-grade **RAG pipeline** using **LangChain**, Pinecone/FAISS, and AWS, reducing LLM hallucinations by 32% and improving retrieval latency by 45% across 1M+ documents.
- Fine-tuned 7B/13B **Qwen3-VL** LLMs using **SFT** and **QLoRA** on **AWS SageMaker** GPU clusters, improving domain-specific accuracy by 21% while cutting inference cost by 60%.
- Deployed production **LLM serving infrastructure** on **GCP Vertex AI** with **vLLM** and TensorRT-LLM, serving 13B-parameter models at sub-200ms latency with 40% cost reduction for 500K+ requests/day.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow**, Terraform, and Vertex AI Pipelines, including CI/CD-driven deployment pipelines; led platform modernization for 15+ ML teams, compressing deployment cycle from 3 weeks to 2 days.
- Deployed multi-agent conversational systems using **LangChain Agents**, Streamlit UIs, and AWS Bedrock API gateways with **event-driven, async processing**; mentored engineering teams on RAG scaling and context-window optimization.
- Built **observability and real-time dashboarding** infrastructure (Grafana, Power BI, Looker) to track agent/system health, operational metrics, and model drift.
- Built Aadhaar-Voter ID linking UI in **React**, scaling to handle high-traffic production loads with resilient, reusable components.
- Migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78%.
- Developed an editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.
- Led a federated design system adopted by 50+ teams and enforced UI standards.
- Integrated Web Vitals + **Sentry** for tracing/observability of LCP, FCP, TTI across 500K+ sessions/month.
- Built a secure file uploader with presigned S3, retry/resume, chunked upload, and Fetch API.
- Implemented keyboard navigation, focus management, ARIA roles, and screen-reader support using semantic HTML.
- Built a schema-to-React renderer supporting dynamic layouts, conditional fields, async data bindings, and versioned JSON configs, cutting UI rollout time.

### Software Engineer — Shiv Technolabs

Feb 2023 - July 2023

- Leveraged deep knowledge of browser internals to optimize performance, reducing reflow, repaint, and bundle sizes to improve Core Web Vitals.
- Implemented advanced state management patterns across complex application workflows using MobX and modern ES6+ JavaScript.
- Led frontend performance audits, analyzing JS engine compilation behavior and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Led the transition to **Next.js**, integrating **RESTful APIs** for seamless data transfer, decreasing bounce rate by 15% and growing traffic.

## Software Engineer — Sourcebae

May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping **REST API** backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent Agile product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built an offline-first UI in React Native using SQLite for local data storage and background sync to update the server when reconnected.

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## PROJECTS

### TradeBrains — Automated Real-Time Algorithmic Trading Platform

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF** (**SFT**, **Reward Model**, **PPO**) and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

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## EDUCATION

### Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75

August 2016 - May 2020

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## ACHIEVEMENTS

- [AWS Certified Developer Associate](#).
- [AWS Certified AI Practitioner](#).
- [AWS Certified Cloud Practitioner](#).
- [Hacker Rank Javascript Basic](#).
- [Hacker Rank CSS certification](#).
- [Sololearn React + Redux](#)

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## TECHNICAL SKILLS

- **LLMs & Agents:** LangChain, LangGraph, ReAct reasoning/action loops, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, MCP Servers, Agent Orchestration & Tooling, RAG Systems, Prompt Engineering
- **Training & Optimization:** LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization
- **Data Engineering:** Apache Spark (PySpark, Spark SQL), Databricks (Lakehouse, Mosaic AI), Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Feature Store (Feast)
- **MLOps & Infrastructure:** MLflow, Kubeflow, Kubernetes, Docker, Terraform, CI/CD Pipelines, Model Monitoring, Observability & Tracing, Drift Detection, A/B Testing
- **Cloud:** AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery)
- **Backend / Middleware Engineering:** Python, FastAPI, Node.js, Go, TypeScript, REST APIs, GraphQL, gRPC, Microservices, WebSocket, Webhooks, Event-Driven Architecture, Redis, n8n
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **Frontend Frameworks:** React.js, Redux/RTK, Next.js, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools & Practices:** Azure, Docker, Webpack, Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase, Agile/Scrum
- **Coursework:** System Design, Database Management Systems, Data Structures and Algorithms, Object Oriented Programming
- **Other:** RESTful Web Services, Integration Testing, Micro-Frontends